

Hide menu

Introduction

Introduction to Linear Models

Reading the Data

Implementing Linear Models for Image Classification

Neural Networks and Deep Neural Networks for Image Classification

Deep Neural Networks with Dropout and Batch Normalization

Introduction

Convolutional Neural Networks

Understanding Convolutions

CNN Model Parameters

Working with Pooling Layers

Implementing CNNs on Vertex AI by using a pre-built TF container

Video: Implementing CNNs on Vertex AI by using a pre-built TF container

Video: Lab Introduction - Classifying images with pre-built TF Container on Vertex AI

Graded App Item: Lab: Classifying Images with pre-built TF Container on Vertex AI

Reading: Readings

Quiz: Quiz

Introduction

Preprocessing the image data

Model parameters and the data scarcity problem

Data Augmentation

Transfer Learning

Summary

Quiz

Submit your assignment

Due Feb 11, 11:59 PM IST

Receive grade

To Pass 80% or higher

Like Dislike Report an issue

Congratulations! You passed!

Grade received 100%

Latest Submission Grade 100%

To pass 80% or higher

Go to next item

		Try again
1.	Which factor does not affect the accuracy of the deep neural network, or DNN? ☐ Inadequate data ☐ Transfer function ☒ Pixel randomization ☐ Network architecture Correct Correct. The accuracy of the deep neural network, or DNN, is not affected by the pixel randomization because the data is not structured hierarchically.	1 / 1 point Your grade100% View FeedbackWe keep your highest score
2.	How many learnable parameters does a pooling layer have? ☐ One ☒ Zero ☐ Two ☐ Four Correct Correct. The pooling layer doesn't have learnable parameters because it only calculates a specific number. Thus, the number of parameters in this layer is zero.	1 / 1 point
3.	What kind of padding methods are available in Keras? ☒ Same padding and valid padding ☐ Casual padding and valid padding ☐ Same padding and casual padding ☐ Casual padding Correct Correct. Keras has two padding methods available: one is called "same" and the other is called "valid." In general, when small, square, and odd-numbered sizes are used for the kernels, the differences are not very meaningful. Also, Keras provides built-in support for padding.	1 / 1 point
4.	Which CNN model parameter helps to maintain the same size across the input and the output of the convolutional layer? ☐ Kernel size ☐ Input channels ☐ Strides ☒ Padding Correct Correct. Padding extends the area of an image in which a convolutional neural network processes. The approach adds a border around the input values in the original image. Therefore, it helps you maintain the same size across the input and the output of the convolutional layer.	1 / 1 point
5.	What is a kernel in a convolutional neural network? ☐ Kernels are the size of the step by which the filter slides across the input image. ☒ Kernels are the building blocks of CNNs because they are used to extract the right and relevant features from the input data using the convolution operation. ☐ A kernel is the area of an image in which a convolutional neural network processes. ☐ A kernel is a parameter that depends on the number of channels in the input image. Correct Correct. A kernel is only a filter that is used to extract the features from the images.	1 / 1 point
6.	What is convolution? ☐ Convolution is a table that is often used to describe the performance of a classification model (or "classifier") on a set of test data for which the true values are known. ☐ Convolution is a function that is used to reduce the spatial size of a representation and increase the number of parameters and amount of computation in a network. ☒ Convolution is the process of "sliding" a kernel across an image. ☐ Convolution is a parameter that helps to maintain the same size across the input and the output. Correct Correct. A convolution is the mathematical combination of two functions to produce a third function.	1 / 1 point
7.	What does the max-pooling operation do in a convolutional neural network? ☒ It returns the maximum value out of all the input data values passed to a kernel. ☐ It returns the average value out of all the input data values passed to a kernel. ☐ It calculates the ratio for each patch of the feature map. ☐ It returns the minimum value out of all the input data values passed to a kernel. Correct Correct. A pooling layer that relies on max-pooling does not require any weights, because the operation only cares about the largest of the input values evaluated by the kernel. This means that during training none of the parameters of the pooling layer need to change.	1 / 1 point
8.	What is true about strides? (Choose two answers) ☐ Stride refers to the number of pixels by which the input matrix slides over the filter matrix. ☐ Larger strides will produce a larger feature map. ☒ Using a stride with a value greater than 1 will reduce the shape produced by the convolutional layer. Correct Correct.The size of the output will be divided along every dimension by the size of the stride step. ☒ Strides are the size of the step by which the filter slides across the input image. Correct Correct. Using a larger step will skip input pixels and produce fewer output values.	1 / 1 point

